Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	PMAT1/N05

Past Paper 5

Time: 25 minutes | Paper Reference **PMAT1/N05**

Mathematics

Level 1
Section A (Non-Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
14 /14

My signature confirms that I will not discuss the content of the test with anyone.
Master Functional Skills (MFS)
Signature: _____

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagram are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used**.
- Take the value of π to be 3.14

Information

- The total mark for this section is 14.
- The marks for **each** question are shown in brackets
  – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ▶

P67366A

©2020 Pearson Education Ltd.

1/1/1

P 6 7 3 6 6 A 0 1 0 8

Pearson

MASTER FUNCTIONAL SKILLS

ONLINE LEARNING PLATFORM & ACCREDITED COURSES

Pearson Edexcel & Open Awards Accredited Suite · Entry 3, Level 1 & Level 2

Pass Your Functional Skills Exams With Confidence

100% Pass Guarantee | Dyslexia-Friendly Learning | 1-to-1 Expert Tutor Support

Master Functional Skills provides an adaptive, dyslexia-friendly online learning platform tailored for adult learners, apprentices, and NHS healthcare staff. Everything you need to revise, practice, and pass is at your fingertips.

1. Smart Diagnostic Initial Assessment

Calculate your exact working level and pinpoint knowledge gaps instantly

- **Exact Grade Calculation:** Adaptive assessments calculate your exact level across all topics.
- **Personalised Study Pathway:** Skip what you know; focus 100% on the topics needed to pass.
- **Real-Time Exam Readiness:** Predictive tracker confirms when you are 100% ready to book.

2. Comprehensive Video Lessons & Worked Solutions

Bite-sized visual explainers covering every single Maths & English curriculum standard

- **Step-by-Step Video Lessons:** Master percentages, algebra, perimeter, ratio, and reverse fractions.
- **Multi-Ink Handwriting Models:** Visual 3-colour methods showing working, carries, and examiner tips.
- **Tutor Shortcuts & Golden Rules:** Proven non-calculator mental arithmetic hacks and BIDMAS tricks.

3. Full Mock Exams & Real-Time Performance Analytics

Practice in exam conditions and watch your scores and confidence climb

- **Authentic Exam Papers:** Timed Section A & Section B papers formatted exactly like Pearson Edexcel.
- **Instant Topic Breakdown:** View percentage scores across Number, Measure/Shape, and Data.
- **Historical Score Tracking:** Monitor past attempts to ensure continuous improvement before exam day.

Ready to Master Your Functional Skills & Pass First Time?

Join thousands of healthcare staff, apprentices, and adult learners across the UK.

Instant 24/7 access on desktop, tablet, and mobile with no software to install.

START LEARNING TODAY AT

MasterFunctionalSkills.co.uk

SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1 Here are four numbers.

1.878 1.76 1.7 1.703

(a) Write these numbers in order of size.
Start with the smallest number.

(1)

Step 1: Pad decimals to 3 decimal places to compare:

- *1.7 = 1.700 (smallest value)*
- *1.703*
- *1.76 = 1.760*
- *1.878 (largest value)*

Order: 1.7, 1.703, 1.76, 1.878 [A1]

1.7, 1.703, 1.76, 1.878 *[A1]*

Here are four fractions.

$\frac{1}{2}$ $\frac{2}{5}$ $\frac{3}{10}$ $\frac{3}{4}$

(b) Write these fractions in order of size.
Start with the smallest fraction.

(2)

Step 1: Convert to common denominator of 20:

- *1/2 = 10/20 (0.50) [M1]*
- *2/5 = 8/20 (0.40)*
- *3/10 = 6/20 (0.30)*
- *3/4 = 15/20 (0.75)*

Step 2: Order from smallest to largest numerator:

- *6/20 < 8/20 < 10/20 < 15/20*
- *Final order: 3/10, 2/5, 1/2, 3/4 [A1]*

3/10, 2/5, 1/2, 3/4 *[A1]*

(Total for Question 1 is 3 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 (a) Work out $10^2 - 2$

(1)

Step 1: Evaluate index power first (10 squared):

- $10^2 = 10 \times 10 = 100$

Step 2: Subtract 2 from 100:

- $100 - 2 = 98$ [A1]

98 [A1]

(b) Work out the range of these numbers.

(2)

−13 −8 −3 −5 −7

Step 1: Identify highest and lowest negative values:

- Highest value = −3 (closest to zero)
- Lowest value = −13 (furthest below zero)

Step 2: Calculate Range = Highest − Lowest:

- Range = $-3 - (-13) = -3 + 13 = 10$ [M1, A1]

Check: Distance on number line is 10 units.

10 [A1]

(Total for Question 2 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3 Vlad needs to put fencing along each side of his garden.
His garden is in the shape of a regular octagon.

Each side of the octagon is 2 m in length.

There will be a 1 m gap in the fencing for a gate.

Fencing costs £22.50 per metre.

Work out the cost of fencing the perimeter of the garden.

(4)

Step 1: Calculate total perimeter of the octagon:

- *8 equal sides of 2 m each*
- *Perimeter = 8 × 2 m = 16 m [M1]*

Step 2: Deduct the 1 m gap left for the gate:

- *Fencing length = 16 m − 1 m = 15 m [M1]*

Step 3: Multiply length by cost per metre (£22.50):

- *Total cost = 15 × £22.50 [M1]*

Step 4: Non-calculator breakdown (10 + 5):

- *10 × £22.50 = £225.00*
- *5 × £22.50 = £112.50*
- *Total = £225 + £112.50 = £337.50 [A1]*

Check: 2250 × 15 = 33750 --> £337.50

£ *337.50* *[A1]*

(Total for Question 3 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4 Kenny is going to borrow £9000 to help buy an electric car.

He pays back the £9000 at 5% simple interest over 4 years.

Kenny thinks he will pay back a total amount of £11 800

Is Kenny correct?
You **must** show your working.

(4)

Step 1: Calculate interest for 1 year (5% of £9000):

- *10% of £9000 = £900*
- *5% = £900 ÷ 2 = £450 per year [M1]*

Step 2: Calculate total interest over 4 years:

- *Total interest = 4 × £450 = £1800 [M1]*

Step 3: Calculate total amount to pay back:

- *Repayment = £9000 + £1800 = £10 800 [M1]*

Step 4: Compare with Kenny's claim (£11 800):

- *£10 800 ≠ £11 800 (£1000 less)*
- *Decision: No, Kenny is not correct. [A1]*

Check: 20% of £9000 = £1800; £10 800 total.

No (£10 800) [A1]

(Total for Question 4 is 4 marks)

TOTAL FOR SECTION A IS 14 MARKS

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	PMAT1/C05

Past Paper 5

Time: 1 hour 30 minutes | Paper Reference **PMAT1/C05**

Mathematics

Level 1
Section B (Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks: 42 /42

My signature confirms that I will not discuss the content of the test with anyone.

Signature: Master Functional Skills (MFS)

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagram are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used**.
- If your calculator does not have a π button take the value of π to be 3.14

Information

- The total mark for this section is 42.
- The total mark for this paper is 56.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ▶

P67367A

©2020 Pearson Education Ltd.

1/1/1/1

P 6 7 3 6 7 A 0 1 1 6

Pearson

SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Wayne earns £13.55 for every hour he works.

These are the hours per day he worked last week.

Day	Wednesday	Thursday	Friday
Hours worked	5 hours	7½ hours	6 hours 30 minutes

How much did Wayne earn last week?

(3)

Step 1: Calculate total hours worked across 3 days:

- *Wednesday = 5 hours*
- *Thursday = 7.5 hours (7½ hours)*
- *Friday = 6.5 hours (6h 30m)*
- ***Total hours = 5 + 7.5 + 6.5 = 19 hours [M1]***

Step 2: Multiply hours by hourly wage (£13.55):

- ***Total earnings = 19 × £13.55 [M1]***
- ***Total earnings = £257.45 [A1]***

Daily check: 67.75 + 101.625 + 88.075 = £257.45

£ **257.45** *[A1]*

(Total for Question 1 is 3 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2 Here is a set of 8 numbers.

184	150	172	168
150	179	175	182

(a) Work out the mean value of this set of numbers.

(3)

Step 1: Calculate sum of all 8 numbers:

- *Sum = 184+150+172+168+150+179+175+182 [M1]*
- *Sum = 1360*

Step 2: Divide sum by count of numbers (8):

- *Mean = 1360 ÷ 8 = 170 [M1, A1]*

170 *[A1]*

☑ (b) Use a reverse calculation to show a check for your answer.

(1)

Reverse check (multiplication):

- *170 × 8 = 1360 [B1]* *[B1]*
- *Verified: Matches original sum of 1360.*

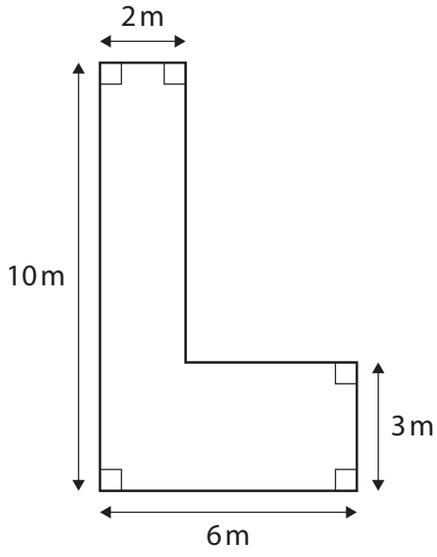
(Total for Question 2 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3 Jamie has this diagram of his kitchen floor.
He is going to cover the floor with tiles.

The tiles are sold in packs.
One pack of tiles will cover 1.2 m^2

Tiles can be cut and joined.



Work out the number of packs Jamie needs to buy.
You **must** show your working.

(5)

Step 1: Calculate missing floor dimensions:

- *Inner horizontal = 6 m − 2 m = 4 m [M1]*
- *Inner vertical = 10 m − 3 m = 7 m*

Step 2: Split L-shape into two rectangles:

- *Left rectangle = 2 m × 10 m = 20 m^2 [M1]*
- *Right rectangle = 4 m × 3 m = 12 m^2*

Step 3: Calculate total floor area:

- *Total Area = 20 m^2 + 12 m^2 = 32 m^2 [M1]*

Check: 2×7 + 6×3 = 14 + 18 = 32 m^2

(Packs calculation continued on next page -->)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Step 4: Calculate packs needed (1.2 m² per pack):

- Packs needed = 32 ÷ 1.2 [M1]
- 32 ÷ 1.2 = 26.666... packs

Step 5: Round up to next whole pack:

- 26 packs = 31.2 m² (not enough coverage)
- 27 packs = 32.4 m² (covers 32 m²)
- Jamie must buy 27 packs. [A1]

Check: 27 × 1.2 m² = 32.4 m² > 32 m²

27 [A1] packs

(Total for Question 3 is 5 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4 Jurita has 103 lamps in her shop to sell.

The price of each lamp is £26.99

Jurita wants to estimate her income from selling these lamps.

Estimate the income for Jurita.
You **must** show your working.

(3)

Step 1: Round values to manageable numbers:

- *103 lamps rounds to 100 lamps [B1]*
- *£26.99 rounds to £27 (or £25 / £30) [B1]*

Step 2: Multiply rounded figures:

- *Estimated income = 100 × £27 [M1]*
- *Estimated income = £2700 [A1]*

Check: 103 × £26.99 = £2779.97 ≈ £2700

£ *2700* *[A1]*

(Total for Question 4 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5 Michael is organising a meeting at his rugby club.
He wants to display the results of the games the rugby team played last year.

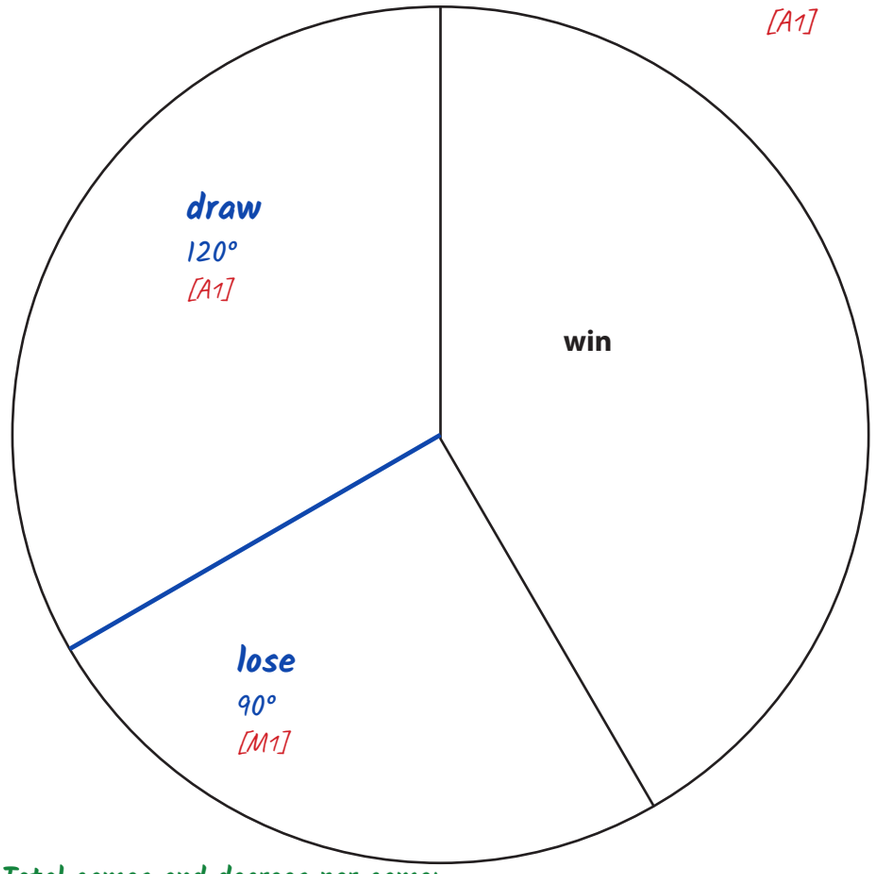
Michael has this information.

result	number of games	angle size
win	25	150
lose	15	
draw	20	

Michael starts to display this information in a pie chart.

Complete the pie chart for Michael.

(3)



Step 1: Total games and degrees per game:

- *Total = 25+15+20 = 60 games; 360° ÷ 60 = 6°/game [M1]*

Step 2: Calculate sector angles for lose and draw:

- *Lose: 15 × 6° = 90° [M1]; Draw: 20 × 6° = 120° [A1]*
- *Check: 150° + 90° + 120° = 360°*

(Total for Question 5 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6 Danielle is a fitness class instructor.
She is updating her website.

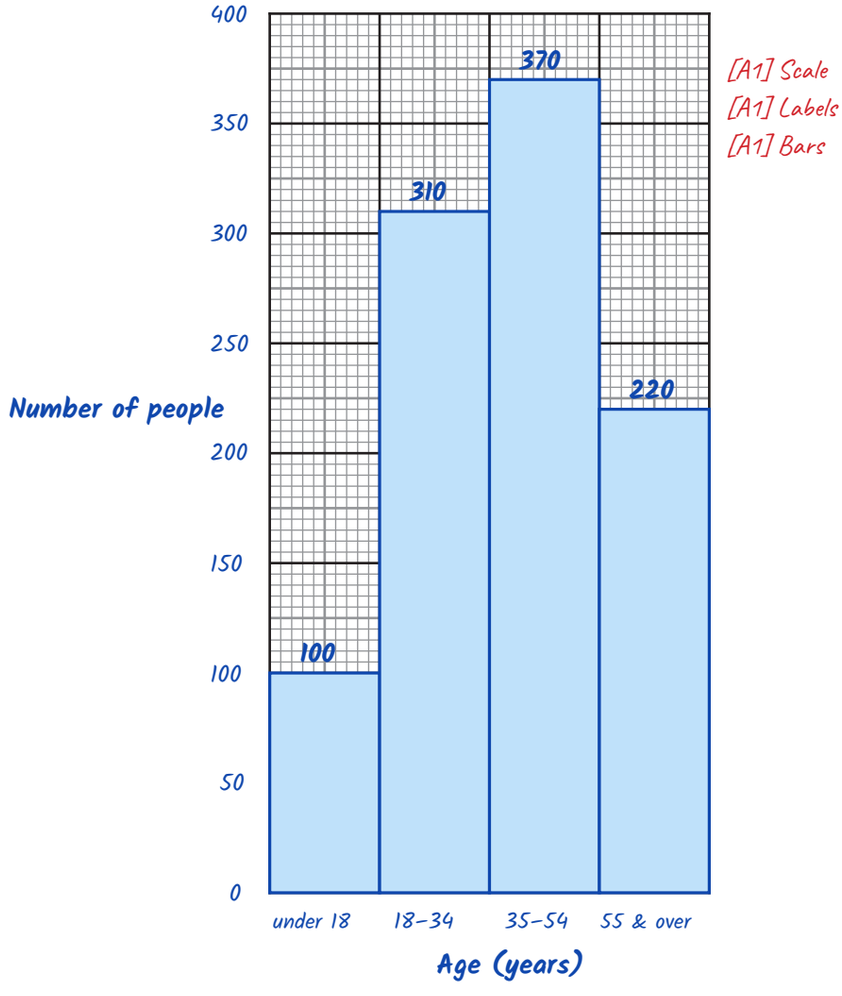
Danielle has this information about the ages of people attending fitness classes in the UK.

age (years)	number of people (for every 1000 people attending classes)
under 18	100
18–34	310
35–54	370
55 and over	220

Danielle wants to display this information as a graph.

(a) On the grid, show a suitable graph for Danielle.

(3)



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

For every 1000 people who attend fitness classes in the UK, 370 are aged 35 to 54

Danielle thinks that 370 out of 1000 is less than 5%

(b) Is Danielle correct?
Show why you think this.

(2)

Step 1: Calculate percentage aged 35–54:

- 370 ÷ 1000 = 0.37
- 0.37 × 100 = 37% [M1]

Step 2: Compare 37% with Danielle's claim of 5%:

- 37% > 5% (far greater than 5%)
- Decision: No, Danielle is not correct. [A1]

No (37% > 5%) [A1]

Danielle collects data about the ages of people who attend her fitness classes.

age (years)	tally	frequency
under 18	𝍸	4
18–34	𝍸 𝍸 ||	12
35–54	𝍸 𝍸 𝍸 |	16
55 and over	𝍸 |||	8

(c) Complete the frequency column for Danielle.

(1)

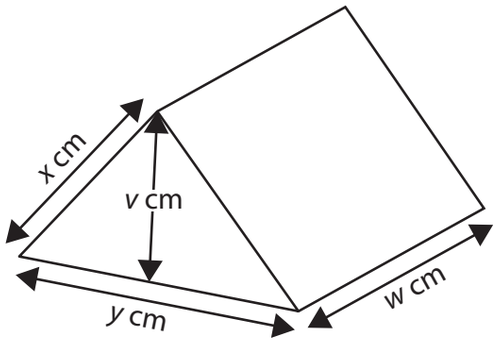
Tally count: 4, 12, 16, 8 (Total = 40) [B1] [B1]

(Total for Question 6 is 6 marks)

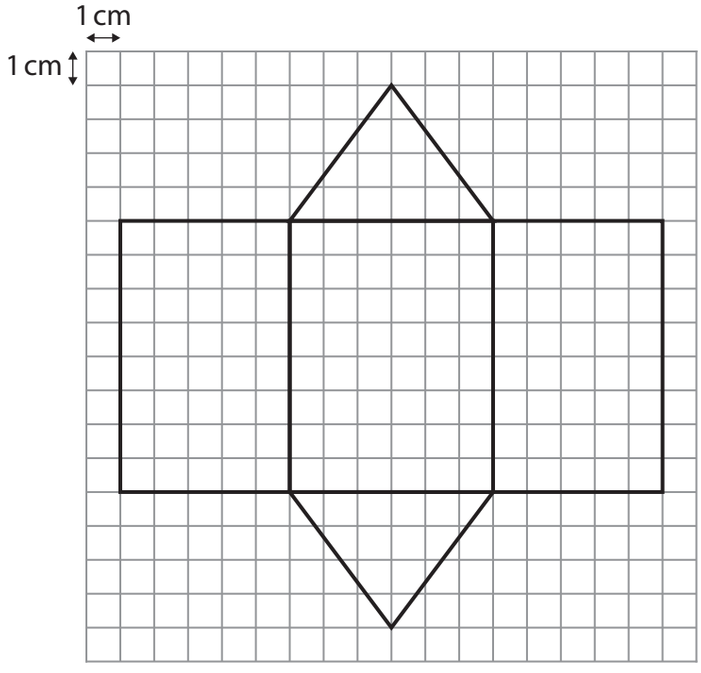
7 Andrew sells cheese.

He wants to put the cheese in a box.
The box is in the shape of a triangular prism.

Here is a sketch of the box.



Here is a net of the box.



Work out the values of *v*, *w*, *x* and *y*.

(3)

Step 1: Read dimensions from 1 cm grid net:

- *Base rectangle: Width = 6 squares --> y = 6 cm*
- *Prism length: Height = 8 squares --> w = 8 cm*
- *Triangle height: 4 squares --> v = 4 cm*
- *Side flaps: Width = 5 squares --> x = 5 cm*

Pythagoras: $3^2 + 4^2 = 25 = 5^2$ (Confirmed)

[A1]

$v =$ **4** cm

$w =$ **8** cm

$x =$ **5** cm

$y =$ **6** cm

(Total for Question 7 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8 Saira employs 24 people.

One third of these people work in the repairs department.

The rest of these people work in the sales department.

How many people work in the sales department?

(3)

Step 1: Calculate staff in repairs (1/3 of 24):

- *Repairs staff = 24 ÷ 3 = 8 people [M1]*

Step 2: Subtract repairs staff from total employees:

- *Sales staff = 24 − 8 = 16 people [A1]*

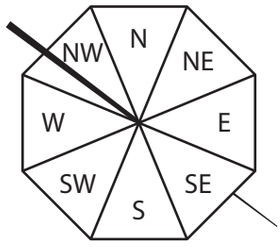
Check: 2/3 of 24 = 8 × 2 = 16 people

16 *[A1]* people

(Total for Question 8 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

9 Here is a fair 8-sided spinner.
Laura spins the spinner once.



(a) Write down the probability that the spinner will land on a section that includes the letter E.

(2)

Step 1: Identify sections with letter E:

- *Total equal sections = 8*
- *Sections with E: NE, E, SE (3 sections) [M1]*

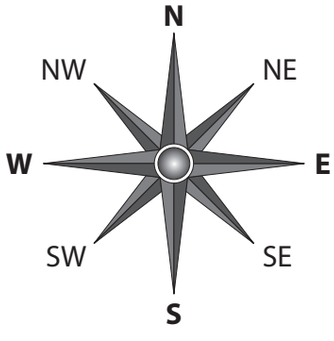
Step 2: Probability = 3/8 [A1]

3

[A1]

8

Here is a compass.



(b) What is the bearing if you walk in the direction southwest?

(2)

Step 1: Bearings measured clockwise from North (000°):

- *South (S) = 180°; West (W) = 270°*
- *SW is halfway between S and W (45° past S)*
- *Bearing = 180° + 45° = 225° [M1, A1]*

225 [A1] °

(Total for Question 9 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10 Polly is designing a display board.

The board needs to

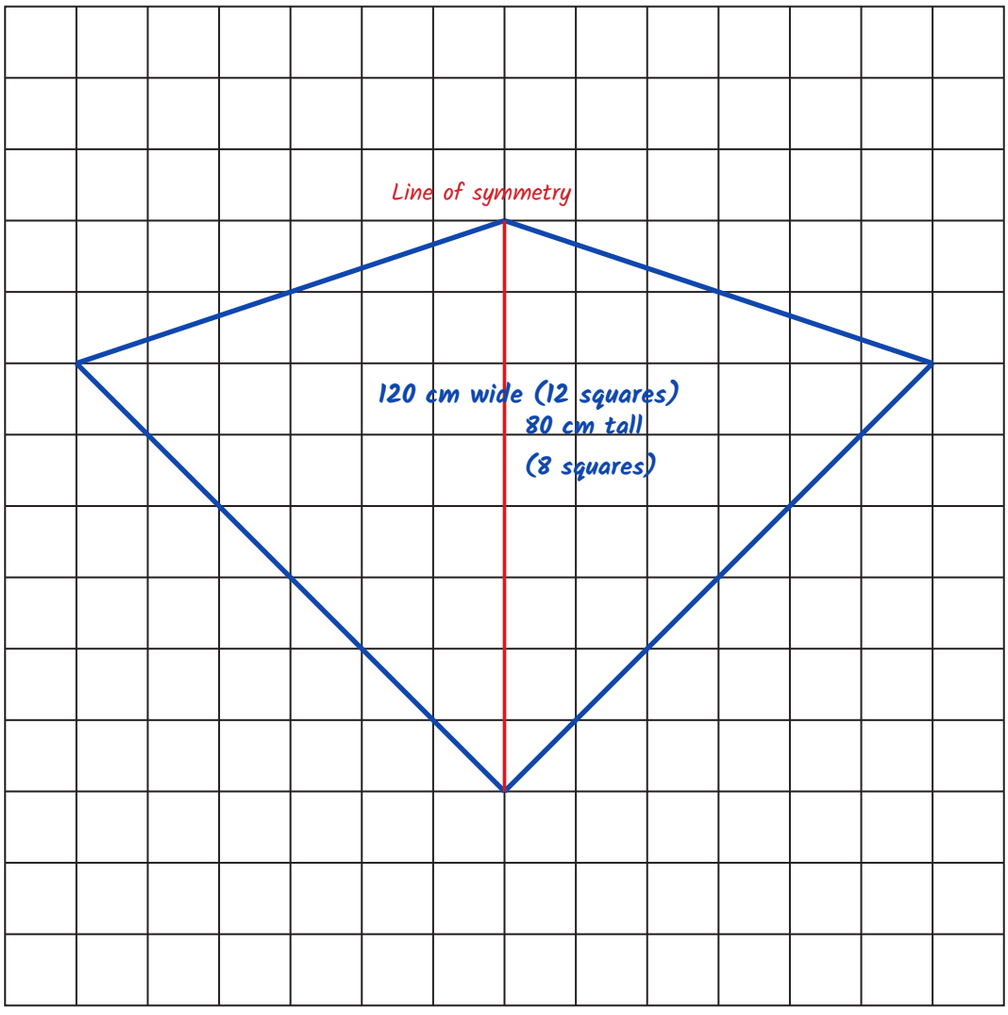
- be in the shape of a kite
- have 1 line of symmetry
- be 120 cm wide
- be 80 cm tall.

Scale: 1 square = 10 cm on board

- Width = 120 cm ÷ 10 = 12 squares [M1]
- Height = 80 cm ÷ 10 = 8 squares [M1]
- Kite with 1 line of symmetry [A1]

Draw a suitable display board on the grid.

(3)



Key 1 square length on the grid represents 10 cm on the board

[A1]

(Total for Question 10 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

11 Natalie wants to buy a travel bag.

She finds the following information about two travel bags.

Holdon bag	Mile bag
Capacity 38.5 litres £82.99	Internal measurements 0.25 m by 0.3 m by 0.6 m £81.99

The Mile bag is in the shape of a cuboid.

Natalie wants to buy the travel bag with the bigger capacity.

She knows that 1 litre = 1000 cm^3

(a) Which travel bag should Natalie buy?
Show why you think this.

(4)

Step 1: Convert dimensions from m to cm (×100):

- *Length = 0.6 m = 60 cm*
- *Width = 0.3 m = 30 cm*
- ***Height = 0.25 m = 25 cm [M1]***

Step 2: Calculate volume in cm^3:

- ***Volume = 60 × 30 × 25 = 45 000 cm^3 [M1]***

Step 3: Convert to litres (1 litre = 1000 cm^3):

- ***Capacity = 45 000 ÷ 1000 = 45 litres [M1]***

Step 4: Compare travel bag capacities:

- *Holdon bag = 38.5 litres; Mile bag = 45 litres*
- *45 litres > 38.5 litres (holds 6.5 litres more)*

Conclusion: Natalie should buy the Mile bag! [A1]

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Mile bag [A1]

(b) Use a reverse calculation to show a check of your answer.

(1)

Reverse check (multiplication):

- 45 litres × 1000 cm³/litre = 45 000 cm³ [B1] [B1]
- 45 000 ÷ 25 ÷ 30 = 60 cm = 0.6 m (Verified)

(Total for Question 11 is 5 marks)

TOTAL FOR SECTION B IS 42 MARKS
TOTAL FOR PAPER IS 56 MARKS

P 6 7 3 6 7 A 0 1 5 1 6

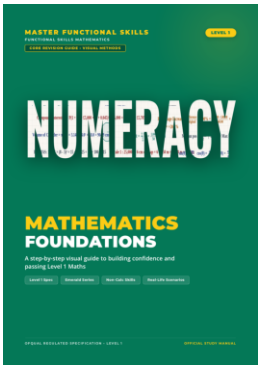
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

MASTER FUNCTIONAL SKILLS STORE

OFFICIAL REVISION GUIDES, WORKBOOKS & FLASHCARDS

Written by accredited specialist tutors to match the exact 2026 Pearson Edexcel standards

Everything You Need To Revise, Practice & Pass Your Maths Level 1 Exam



Core Curriculum Manual

Maths Level 1
Core Study Guide

- Full official curriculum coverage
- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
- Dyslexia-friendly visual format

Spiral Bound Print & PDF

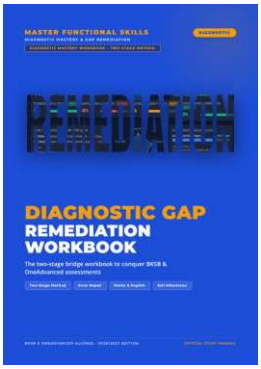


1,000+ Practice Drills

Maths Level 1
Companion Workbook

- 1,000+ exam-style questions
- Progressive difficulty drills
- Full worked step-by-step key
- Targeted fix for weak areas

Spiral Bound Print & PDF



Realistic Mock Papers

Maths Level 1
Official Mock Packs

- Full mock exam paper sets
- Authentic Pearson Edexcel style
- Detailed mark schemes included
- Grade boundaries & pass marks

Exam Booklet Pack & PDF



Pocket Flashcards A6

Maths Level 1
Pocket Revision Cards

- Compact A6 cards for quick study
- Essential formulas & conversions
- Dual-sided question & answer
- High-yield mental arithmetic rules

Boxed Card Deck & Print PDF

Order Official Study Guides & Revision Cards Online

Free Next-Day UK Tracked Shipping · Instant PDF Downloads Included

Trusted by NHS trusts, apprentices & adult learners nationwide.

VISIT THE ONLINE STORE

MasterFunctionalSkills.co.uk